

Technical Contribution – Data Layer & Storage

Project: Serverless Payroll Processing & Payslip Delivery Engine

➤ Overview

As the **Data Layer & Storage Engineer** in a team of three members, I was responsible for designing and implementing the data storage layer of the Serverless Payroll Processing & Payslip Delivery Engine on AWS. My work focused on creating the database schema, preparing payroll datasets, configuring cloud storage, and enabling seamless integration with the payroll processing pipeline.

➤ Responsibilities

- Designed the payroll data model using Amazon DynamoDB.
 - Created and configured all required DynamoDB tables.
 - Populated realistic employee payroll datasets for testing and integration.
 - Configured Amazon S3 storage for generated payslips.
 - Defined the storage hierarchy for payroll documents.
 - Supported the payroll processing workflow by providing a structured and scalable data layer.
 - Prepared the storage layer for payroll history and audit requirements.
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DynamoDB Data Model

I designed and implemented the following DynamoDB tables:

1. employee_master

Stores employee master information required for payroll processing.

(a) Attributes

- employee_id
- employee_name
- email
- department
- designation
- employment_status

- joining_date

(b) Partition Key

- employee_id

Fig 1: employee_master table

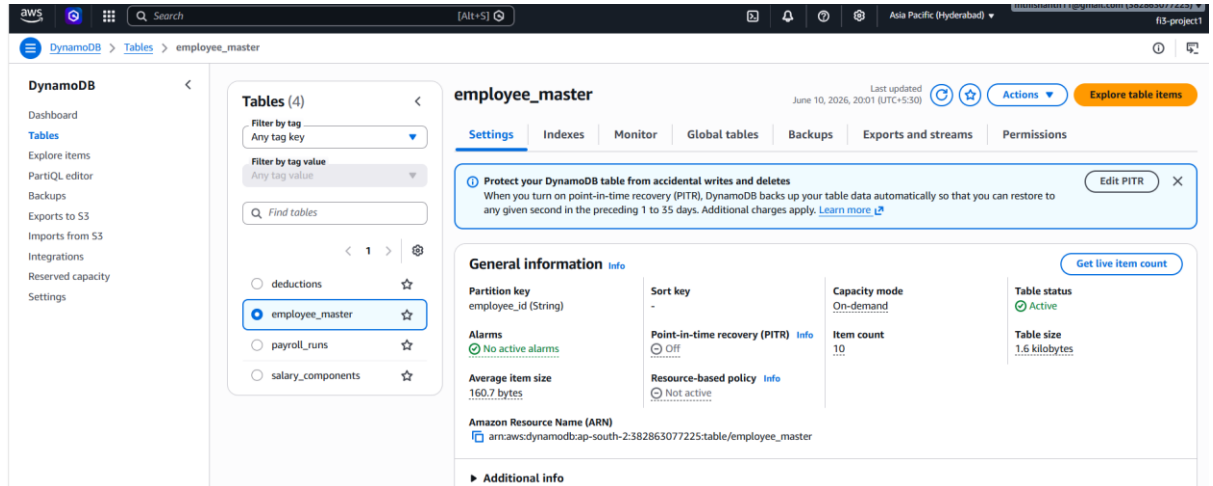


Fig 2: employee_master table (Items)

	employee_id (String)	department	designation	email	employee_na...	employment_status
☐	EMP001	Engineering	Cloud Engineer	swagatasin...	John Doe	ACTIVE
☐	EMP002	HR	HR Manager	swagatasin...	Priya Sharma	ACTIVE
☐	EMP003	Finance	Accountant	swagatasin...	Arjun Patel	ACTIVE
☐	EMP004	Marketing	Marketing An...	swagatasin...	Neha Verma	ACTIVE
☐	EMP005	Operations	Operations M...	swagatasin...	Michael Brown	ACTIVE
☐	EMP006	IT	DevOps Engin...	swagatasin...	Sara Khan	ACTIVE
☐	EMP007	Support	Support Lead	swagatasin...	Ravi Kumar	ACTIVE
☐	EMP008	Finance	Senior Accoun...	swagatasin...	Anita Desai	TERMINATED
☐	EMP009	Engineering	Software Dev...	swagatasin...	Karan Mehta	ON LEAVE
☐	EMP010	Operations	Business Anal...	swagatasin...	Meena Reddy	ON LEAVE

2. salary_components

Stores employee salary structure used for payroll calculations.

(a) Attributes

- employee_id
- base_salary
- hra
- transport_allowance
- special_allowance

(b) Partition Key

- employee_id

Fig 3: salary_components table

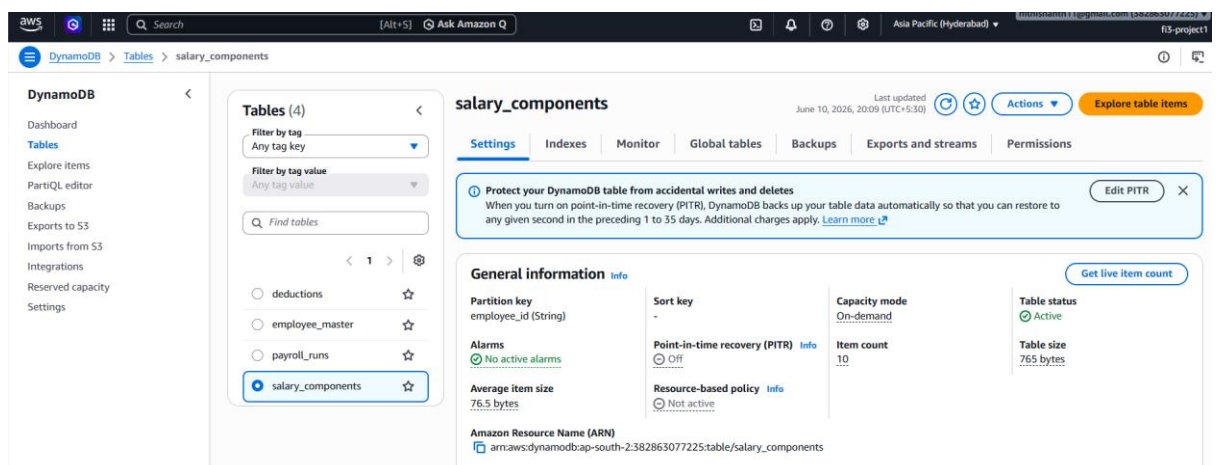


Fig 4: salary_components table (Items)

Completed · Items returned: 10 · Items scanned: 10 · Efficiency: 100% · RCUs consumed: 2						
Table: salary_components - Items returned (10)						
Scan started on June 10, 2026, 20:11:39						
	employee_id (String)	base_salary	hra	special_allowance	transport_allowance	
☐	EMP001	50000	20000	3000	5000	
☐	EMP002	60000	25000	5000	4000	
☐	EMP003	45000	18000	2000	3000	
☐	EMP004	40000	15000	1500	2500	
☐	EMP005	65000	26000	4000	5000	
☐	EMP006	55000	22000	3000	4000	
☐	EMP007	48000	19000	2500	3500	
☐	EMP008	70000	28000	5000	6000	
☐	EMP009	52000	21000	2500	4000	
☐	EMP010	62000	24000	3500	4500	

3. deductions

Stores employee deduction details.

Attributes

- employee_id
- pf_percentage
- professional_tax
- tds_percentage
- loan_deduction

Partition Key

- employee_id

Fig 5: deductions table

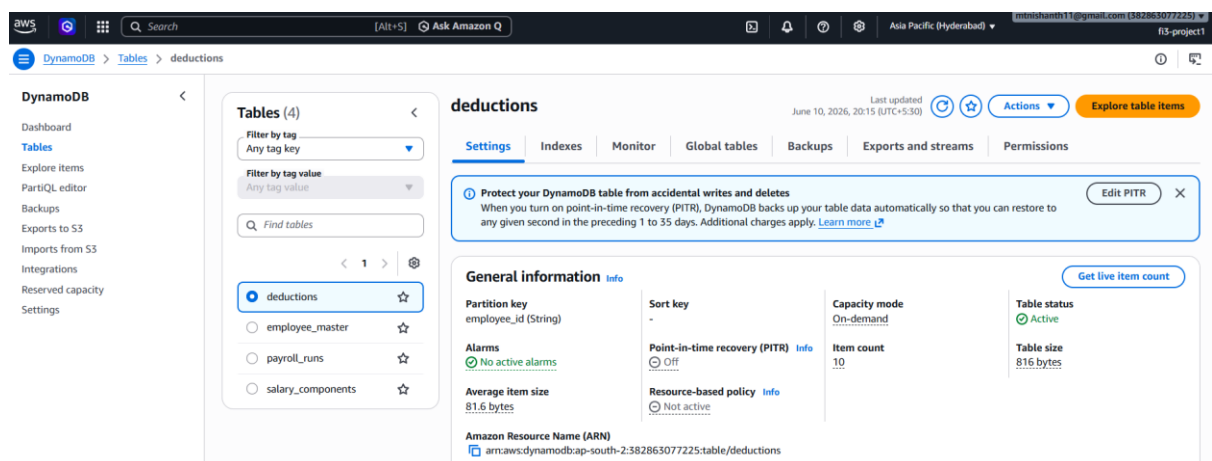


Fig 6: deductions table (Items)

✔ Completed · Items returned: 10 · Items scanned: 10 · Efficiency: 100% · RCUs consumed: 2						
Table: deductions - Items returned (10)						
Scan started on June 10, 2026, 20:17:02						
☐	employee_id (String)	▲	loan_deduction ▼	pf_percentage ▼	professional_tax ▼	tds_percentage ▼
☐	EMP001		2000	12	300	10
☐	EMP002		0	12	300	10
☐	EMP003		1000	12	200	10
☐	EMP004		0	12	200	10
☐	EMP005		1500	12	300	10
☐	EMP006		1000	12	300	10
☐	EMP007		500	12	200	10
☐	EMP008		0	12	300	10
☐	EMP009		0	12	300	10
☐	EMP010		2000	12	300	10

4. payroll_runs

Designed to maintain payroll execution history and audit records generated during payroll processing.

(a) Attributes

- run_id
- employee_id
- month
- gross
- deductions_total
- net_pay
- status
- payslip_s3_key

Fig 7: payroll_runs tables

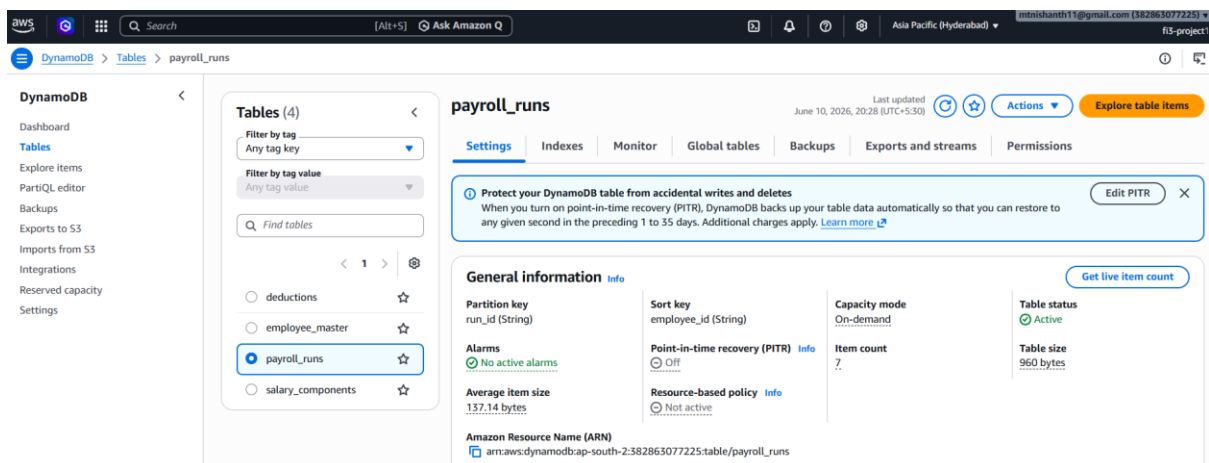


Fig 8: payroll_runs (Returned Items)

Completed · Items returned: 7 · Items scanned: 7 · Efficiency: 100% · RCUs consumed: 2

Table: payroll_runs - Items returned (7)							
Scan started on June 10, 2026, 20:27:04							
☐	run_id (String)	employee_id (String)	deductions_total	gross	month	net_pay	
☐	RUN202606	EMP001	16100	78000	2026-06	61900	
☐	RUN202606	EMP002	16900	94000	2026-06	77100	
☐	RUN202606	EMP003	13400	68000	2026-06	54600	
☐	RUN202606	EMP004	10900	59000	2026-06	48100	
☐	RUN202606	EMP005	19600	100000	2026-06	80400	
☐	RUN202606	EMP006	16300	84000	2026-06	67700	
☐	RUN202606	EMP007	13760	73000	2026-06	59240	

➤ Test Dataset Preparation:

Created sample payroll data for multiple employees representing different employment scenarios, including:

- Active Employees
- On Leave Employees
- Terminated Employees

This dataset enabled validation of payroll business rules and end-to-end integration testing.

Amazon S3 Storage Configuration

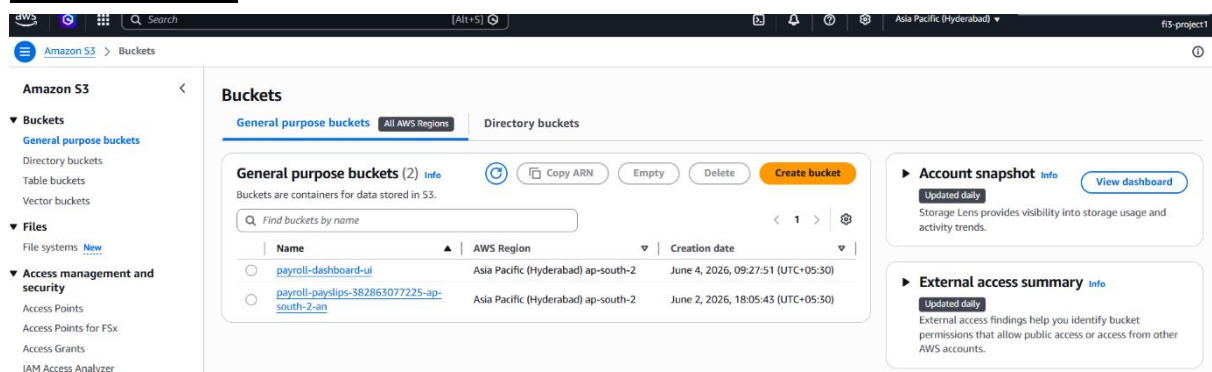
Configured Amazon S3 as the storage layer for generated payroll documents.

Folder hierarchy:

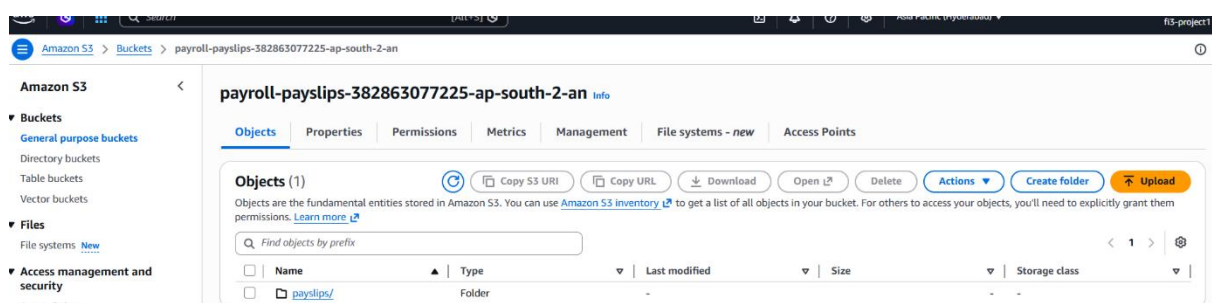
payslips/
└─ 2026/
 └─ 06/
 EMP001.pdf
 EMP002.pdf
 EMP003.pdf

The storage structure supports organized archival and retrieval of employee payslips.

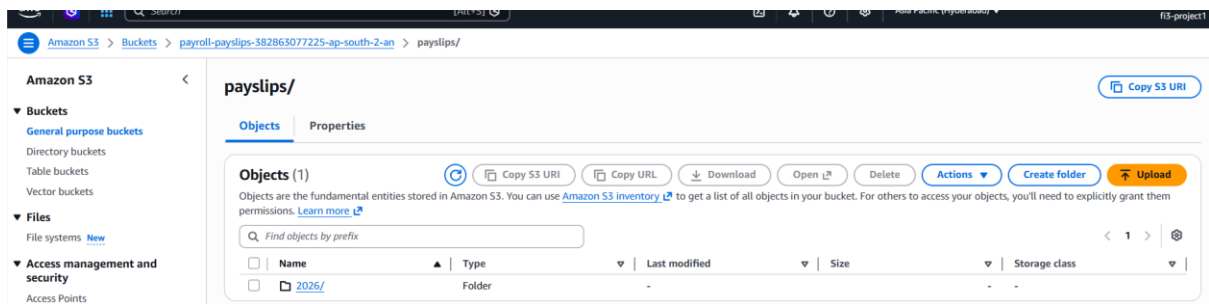
Fig 9(a): Buckets



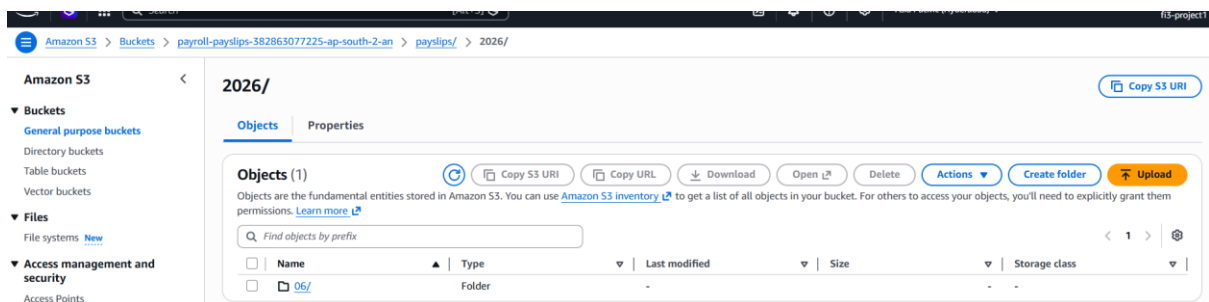
(b)



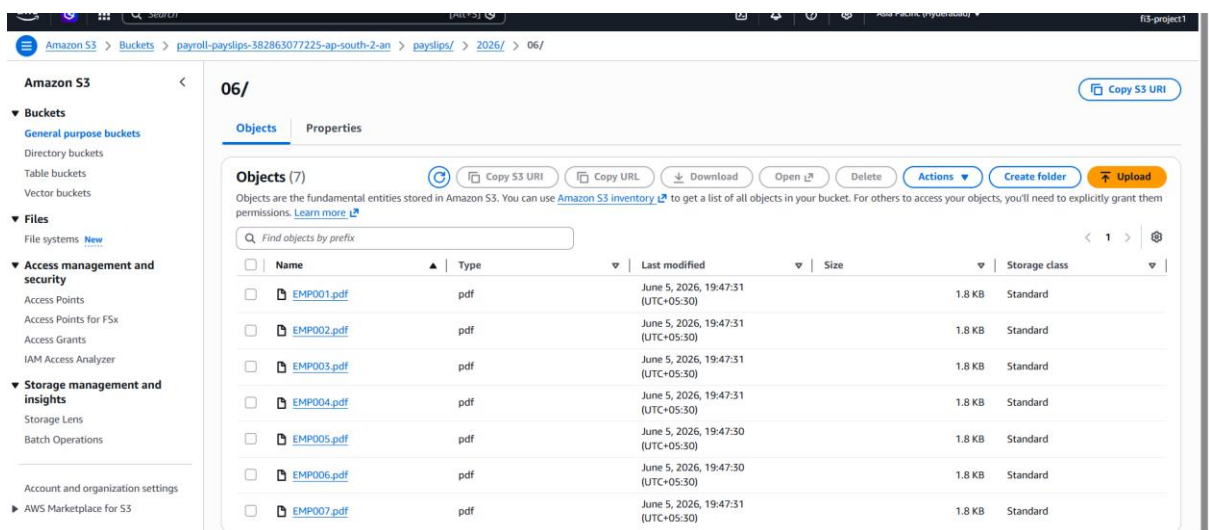
(c)



(d)



(e)



Integration Support:

The data layer was designed to integrate with the serverless payroll workflow:

- Employee information retrieved from employee_master
- Salary details retrieved from salary_components
- Deduction information retrieved from deductions

- Payroll execution results stored in payroll_runs
- Generated payslips stored in Amazon S3

This design enabled seamless interaction with AWS Lambda, Amazon SQS, and Amazon SES during payroll processing.

Key Skills Demonstrated:

- Amazon DynamoDB Data Modeling
 - NoSQL Database Design
 - AWS Cloud Storage (Amazon S3)
 - Serverless Data Architecture
 - Payroll Data Management
 - Data Normalization
 - AWS Console Configuration
 - Cloud Integration Support
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Outcome

The implemented data layer provides a scalable and organized foundation for payroll automation by supporting salary calculations, payroll history management, secure payslip storage, and seamless integration with the serverless processing pipeline. My contribution ensured that the storage architecture was reliable, integration-ready, and aligned with AWS serverless best practices.

Made by:

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